P-C-451E
March 17, 1975
SUPERSEDING
Fed. Spec. P-C-451D
October 23, 1973
(See 6.5)

FEDERAL SPECIFICATION

COATED ABRASIVE, CLOTH, ALUMINUM OXIDE OR SILICON CARBIDE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers coated abrasives comprised of a backing of cloth coated on one surface with aluminum oxide or with silicon carbide abrasive grain.

1.2 Classification. Abrasive cloths covered by this specification shall be of the following types and classes.

a. Type I - Aluminum Oxide
Class 1 - sheets
Class 2 - rolls
Grit sizes - 320, 240, 220, 180, 150, 120, 100, 80, 60, 50, 40, 36, 30, and 24.

b. Type II - Silicon Carbide
Class 1 - sheets
Grit sizes - 320, 240, 220, 180, 150, 120, 100, 80, 60, 50, 40, 36 and 24.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein:

FSC 5350

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Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-3816 - Packaging and Packing of Abrasives and Abrasive Products for Shipment and Storage.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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Department of Commerce Product Standard

PS-8 Grading of Abrasive Grain on Coated Abrasive Products.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Abrasive grain.

3.1.1.1 The abrasive grain for Type I abrasive cloth shall be crushed from titanium toughened, brown abrasive grade aluminum oxide and shall be relatively free of extraneous materials.

3.1.1.2 The abrasive grain for Type II abrasive cloth shall be crushed from abrasive grade silicon carbide and shall be relatively free of extraneous materials.

3.1.1.3 Grading control and tolerance. Grading of abrasive grain shall be in accordance with PS-8 and the tolerances therein shall apply to abrasive grain recovered from the coated abrasive product.

3.1.2 Backing.

3.1.2.1 The backing shall be a drills cloth for grit sizes 60 and coarser and jeans cloth for grit sizes 80 and finer. The cloth shall be dyed and filled preparatory to coating. The greige cloth shall inset the requirements of Table I directly after being conditioned at 70 degrees, plus or minus 2°F and 50 percent plus or minus 2 percent, relative humidity for at least 24 hours.

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TABLE I BACKING REQUIREMENTS

TYPES	GRIT SIZE RANGE	CLOTH TYPE	GREIGE CLOTH YDS/LB (1) ± 5 PERCENT	THREAD COUNT PER INCH (2) WARP	FILLING
I - either of the following:					
	80 and finer	Jeans	3.54	96	64
	80 and finer	Jeans	3.98	84	56
II -	80 and finer	Jeans	3.54	96	64
I - either of the following:					
	60 and coarser	Drills	2.78	72	48
	60 and coarser	Drills	2.78	76	48
II	60 and coarser	Drills	2.78	76	48

(1) Nominal yards per pound of greige cloth is on a 30-inch wide basis.

(2) A minus tolerance of three threads or a plus tolerance of five threads per inch will be permitted in the thread count of the greige goods.

3.1.3 Adhesive

3.1.3.1 The adhesives shall be a good quality, animal hide glue base or equivalent, suitable for the purpose intended.

3.2 Coating. The abrasive cloth shall be closed coat. The abrasive grain shall be applied to one side of the cloth backing only and shall completely cover the surface of the backing.

3.3 Tensile breaking strength. The minimum average breaking strength of the coated abrasive cloth shall be in accordance with Table II when tested as described in 4.4.2.

TABLE II - TENSILE BREAKING STRENGTH

Types and Classes	Backing Desig.	Grit Size Range	Pounds/Inch - Minimum Average Machine Direction	Cross Machine Direction
Type I & II Class 1	Jeans	80 and finer	100	45
Type I, Class 1	Drills	60 and coarser	130	50
Type I, Class 2	Jeans	80 and finer	100	-
Type I, Class 2	Drills	60 and coarser	130	-
Type II, Class 1	Drills	60 and coarser	145	40

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3.4 Form and sheet dimensions.

3.4.1 Sheet. Unless otherwise specified (see 6.2) class 1 abrasive cloth shall be furnished in 9 X 10 or 9 X 11 inch sheets, with a tolerance of plus or minus 1/8 inch in the machine direction and plus or minus 1/16 inch in the cross machine direction when measured at and after conditioning at 70°F plus or minus 2°F, and 50 percent plus or minus 2 percent relative humidity for at least 24 hours.

3.4.2 Roll. Class 2 abrasive cloth shall be furnished in rolls with a standard length of 50 yards with a tolerance of minus zero, plus 12 inches and in the nominal widths specified (see 6.2). There shall be no more than two breaks in each 50 yard roll.

3.5 Identification marking. Each sheet of class 1, Types I and II abrasive cloth shall be marked with the manufacturer's name or trademark (of such character that the source may be readily determined) and the grit size. (See 3.5.1). Each roll of Class 2, Type I abrasive cloth shall be marked with the manufacturer's name or trademark, and the grit size. (See 3.5.2). All markings are to be legible and permanent.

3.5.1 Sheets. 9 X 10, or 9 X 11 inch must be marked as in 3.5.

3.5.2 Rolls. Rolls 4 1/2 inch and wider shall be marked as in 3.5 at least once every yard.

3.5.3 Sheets or rolls, smaller than the dimensions shown may be furnished with partial marking on each item.

3.6 Workmanship. The workmanship shall be of the equal or better in every respect to that considered good practice by industry. The abrasive cloth shall be free from imperfections affecting its appearance or major defects which may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. The sample size levels of inspection shall be in accordance with MIL-STD-105.

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4.2.1 Lot. All abrasive cloth of the same type, class and grit size manufactured under essentially the same conditions and offered for delivery at one time shall be considered a lot for the purpose of examination and tests.

4.2.2 Sample. A random sample of filled packages shall be selected from each lot of filled packages offered for delivery, at Inspection Level I.

4.2.3 Sampling for visual and dimensional examination of the abrasive cloth. Five - sheets shall be selected at random from each sample package selected in accordance with 4.2.2, each sheet from a separate folder.

a. On Class 2 abrasive cloth, in roll form, approximately 15 feet shall be unrolled from each sample roll selected from the packages selected in 4.2.2 for the visual and dimensional examinations.

4.2.4 Sampling for tests. Sample sheets may be selected from those selected in accordance with 4.2.3.

4.3 Examination and tests. The acceptable quality levels (AQL's) shall be in accordance with MIL-STD-105.

4.3.1 Examination.

4.3.1.1 Packaged and packed abrasive cloth. Abrasive cloth ready for shipment, selected in accordance with 4.2.2, shall be examined with AQL's and classification of defects in accordance with MIL-P-3816.

4.3.1.2 Individual sheets and rolls. Samples of abrasive cloth selected in accordance with 4.2.3 shall be examined in accordance with Table III. The AQL's for defects listed in Table III shall be 2.5 percent defective.

4.3.1.3 The sheets and rolls shall be examined for conformance to size (see 3.4), marking, abrasive grain coverage, and various elements of workmanship. The abrasive coated surface shall be examined for defects, such as foreign matter, unevenness of or voids in the coating, loose grain, backing cracks, or other defects which may impair (major defects) the usefulness of the abrasive cloth.

TABLE III - CLASSIFICATION OF DEFECTS (END ITEM)

Examine Applicable Paragraph	Defects	Major	Minor
1.2	Grit size of abrasive not as specified or ordered	X	
3.1.1.1 4.3.1.2	Excessive extraneous material present in abrasive	X	
3.2 4.3.1.2	Coating not closed as specified	X	
3.4.1	Dimensions of sheets, not correct size or not within tolerances.	X	
3.4.2	Size of rolls not as specified	X	

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TABLE III - CLASSIFICATION OF DEFECTS (END ITEM) (Continued)

Examine Applicable Paragraph	Defects	Major	Minor
3.5	Identification marking omitted, illegible, incomplete or incorrect.		X
3.6 4.3.1.2	Sheets and rolls torn or cracked.	X	

4.3.2 Lot acceptance tests. Samples selected in accordance with 4.2.2 shall be subjected to the tests in 4.4, AQL 2.5 percent defective.

4.4 Tests.

4.4.1 Abrasive grit size.

4.4.1.1 Grading for grit size. Grading to determine the grit size, including recovery of abrasive grain from the coated abrasive product, preparation of grain for grading, and procedures shall be in accordance with PS-8.

4.4.1.2 Certification of grading. Grading tests specified in 4.4.1.1 may be omitted in instances where the contractor produces certification in writing that the abrasive grain complies with the requirements of PS-8 for aluminum oxide on Type I, and silicon carbide on Type II material. Contractors are urged to include this certification with material offered for inspection and inspectors are urged to determine whether such certification is included with such material in order to preclude the necessity for grading tests.

4.4.2 Breaking strength. The tensile breaking strength of the coated abrasive directly after conditioning for at least 24 hours at 70° plus or minus 2°F. and 50 plus or minus 2 percent relative humidity, shall be determined by method 5102 of Federal Test Method Std. No. 191 except that the jaws of the testing machine shall be 5 inches apart, and that six specimens 1 inch wide shall be taken both longitudinally (machine direction) and transversely (cross machine direction) from sample sheets and rolls. The average tensile strength in both directions shall be computed from at least three specimens. For rolls less than 7" wide the average tensile strength need only be computed in the machine direction. Specimens breaking at the jaws of the tensile testing machine should be discarded. (See 3.3 and Table II.)

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing of abrasive cloth shall be in accordance with MIL-P-3816, as specified for level A, B, or C (see 6.2). Unless otherwise specified, the number of sheets per folder will be 50 unless otherwise specified.

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5.2 Marking

5.2.1 Civil agencies. In addition to any special markings required by the contract or order, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to any special markings required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Type I. Type I aluminum oxide cloth is generally intended for use in abrading metal of high tensile strength.

6.1.2 Type II. Type II silicon carbide cloth is generally intended for use in abrading metals of low tensile strength.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents (if applicable).

a. Title, number and date of the specification.

b. Type, class, and grit size required (see 1.2).

c. Forms and dimensions required.

d. Selection of applicable levels of preservation, packaging and packing required (see 5.1).

e. Number of sheets per folder or rolls per package.

f. Special marking, if required.

6.3 Silicon Carbide. A synthetic mineral, crystalline in structure, chemically SiC having a Mohs scale hardness of 9.6, a Knoop hardness of 2480 approximately and a specific gravity of 3.2.

a. Aluminum Oxide. A synthetic mineral, crystalline in structure with titania additives to impart toughness, having a Mohs scale hardness of 9.4, a Knoop hardness of 2050, approximately and a specific gravity of 3.96.

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6.4 Work value. Coated Abrasives should be stored in a cool, even temperature area where the relative humidity is somewhere between 40 and 50 percent. Do not store on concrete floors. The best work value will be obtained if stored and worked in areas having a relative humidity of approximately 50 percent.

6.5 Supersession. P-C-451E supersedes P-C-451D (which was erroneously circulated as a coordinated document) and P-C-451C, dated April 24, 1967.

Custodian:

Army - MR
Navy - SR
Air Force - 84

User activities:

Army - MI, ME
Navy - YD, MC

Preparing activity:

Air Force - 84

Project No. 5350-0053

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION P-C-451E Coated Abrasive, Cloth, Aluminum Oxide or Silicon Carbide

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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